

A Three-Tier Knowledge Graph RAG for Vietnamese Legal Question Answering

Hong Hien LE ^{a,1}, Dinh Hien NGUYEN ^a, Quoc Hung NGO ^b and Viet Dung DANG ^a

^a*Faculty of Computer Science, University of Information Technology, Vietnam National University Ho Chi Minh City, Vietnam*

^b*School of Computer Science, University College Dublin, Ireland*

Abstract. To answer legal questions accurately, a system must navigate hierarchical document structures, follow cross-references, and reason across multiple hops, but standard Retrieval Augmented Generation (RAG) systems cannot do this. Vietnamese legal text makes this gap worse. Tonal language, domain-specific abbreviations, and a rigid structural hierarchy (Law→Chapter→Article→Clause→Point) all compound retrieval difficulty. We propose a three-tier knowledge graph RAG architecture that extends the Legal-Onto Rela-model with a formal knowledge model $\mathcal{M} = (\mathcal{C}, \mathcal{G}, \mathcal{T}, \Phi)$, integrating an OWL ontology for terminological knowledge, a domain-specific knowledge graph for assertional knowledge, and a document tree forest for structural navigation. The three retrieval tiers, specifically LLM-guided tree traversal, dual-level semantic retrieval with intent-aware Personalized PageRank, and Reciprocal Rank Fusion, operate over these representations. To evaluate the method, we conducted experiments on 200 legal questions in Vietnam belonging to two different legal fields: enterprise law (150 questions on company formation, capital, dissolution) and traffic law (50 questions on violations, penalties, procedures) across 13 legal documents. Our system achieves 84.5% Hit@10 (MRR 0.461), outperforming TF-IDF (40.5%) and BM25 (36.0%) by substantial margins. Per-domain results confirm generalization: 83.3% Hit@10 on enterprise law and 88.0% on traffic law, showing the flexibility and strong querying capabilities across various legal domains. We are not aware of prior work combining ontology-enhanced knowledge graphs with multi-tier retrieval for Vietnamese legal question answering.

Keywords. Knowledge Graph, Legal Question Answering, Retrieval Augmented Generation, Ontology, Vietnamese NLP, Multi-Tier Retrieval, Multi-Domain

1. Introduction

Legal forums and social networks in Vietnam are flooded with questions on business formation, traffic penalties, and regulatory compliance. Yet answers are often fragmented, contradictory, or cite repealed provisions.

Legal documents exhibit **complex hierarchical structures**: a single law organizes content into parts, chapters, sections, articles, clauses, and points, where each level carries distinct legal scope. They also contain pervasive **cross-references**: answering “If a member borrows money to meet charter capital and later withdraws it, what are the

¹Corresponding Author: Le Hong Hien, Faculty of Computer Science, University of Information Technology, Vietnam National University Ho Chi Minh City, Vietnam; E-mail: hienlh.16@grad.uit.edu.vn.

penalties?” requires chaining Article 47 (member obligations) to Article 16, Clause 5 (prohibited acts) across different chapters.

The **Vietnamese legal domain** adds further complexity. Vietnamese is tonal with ambiguous compound word boundaries. Legal practitioners use extensive abbreviations (*HDQT* for “Hoi dong quan tri”, *CTCP* for “Cong ty co phan”) that general models miss. Vietnamese law follows a rigid hierarchy such as Constitution, Codes, Decrees, and Circulars, where document type determines legal authority.

Traditional RAG systems [1] retrieve passages by vector similarity between query and document embeddings [2]. This works adequately for many domains [3,4], but legal QA exposes its weaknesses. Legal terminology carries precise meanings that general embeddings blur (**semantic gap**); documents organize content hierarchically, yet vector search treats them as flat text (**no structural awareness**); and answering a question often requires chaining references across articles, something dense retrieval cannot do (**no multi-hop reasoning**).

Existing approaches tackle parts of this problem. LightRAG [5] and GraphRAG [6] model entity relationships but disregard document hierarchy. PageIndex [7] exploits structure but has no semantic reasoning component. Legal ontology systems [8,9] formalize domain knowledge but remain disconnected from modern retrieval pipelines. We propose a **three-tier knowledge graph RAG** architecture that brings these capabilities together:

1. **LLM-guided tree traversal** through document hierarchies for structural navigation (Tier 1),
2. **Dual-level semantic retrieval** with intent-aware Personalized PageRank on a domain knowledge graph (Tier 2),
3. **Reciprocal Rank Fusion** for cross-tier merging with knowledge graph expansion (Tier 3).

Our contributions: (1) a three-tier RAG architecture combining structural, semantic, and graph-based retrieval; (2) a Vietnamese legal KG (5,936 entities, 5,633 relations) with semi-automatic construction; (3) a multi-intent PPR algorithm adapting edge weights to query intent. Experiments on 200 questions across two legal domains show 84.5% Hit@10 (+44.0pp over the best baseline).

2. Related Work

RAG and KG-augmented retrieval. RAG [1] grounds generation in retrieved evidence. Since early dense retrieval pipelines [10], designs have grown more sophisticated: Self-RAG [11] adds reflection, CRAG [12] corrective retrieval, and HyDE [13] hypothetical document generation; surveys [3,4] trace this progression. On the knowledge graph side, GraphRAG [6] enables multi-hop reasoning and LightRAG [5] offers lighter entity-relation extraction, but both treat documents as bags of entities and lose structural context. Embedding-based KG methods (TransE [14], RotatE [15]) capture relational patterns yet are limited to single-hop prediction. Personalized PageRank [16,17] computes query-specific node importance. PageIndex [7] takes a different path with LLM-guided tree traversal, achieving high precision with few articles, but missing content outside the traversal path.

Legal NLP. Legal IR has moved from lexical methods [18,19] toward domain-adapted transformers such as LEGAL-BERT [20], which capture legal phrasing bet-

ter than general models but still treat documents as flat passages. Ontology-based approaches [9] add richer structure; Legal-Onto [8] combines the Rela-model [21] with conceptual graphs for Vietnamese Land Law, though it relies on manual inference rules and was not designed for retrieval. For Vietnamese specifically, PhoBERT [22] and Vn-CoreNLP [23] provide foundational NLP capabilities, and Ngo et al. [24] combined BM25 with BERT [25] and domain heuristics, an effective baseline that nonetheless lacks graph-based reasoning. Surveys [26,27] confirm that bridging formal legal knowledge with scalable retrieval remains an open problem.

Positioning. Table 1 maps each system’s capabilities. The gap motivating our work is the absence of a system that combines ontological knowledge representation with multi-tier retrieval for Vietnamese legal text.

Table 1. Feature comparison of legal QA approaches. *Not a RAG system; included for ontology comparison.

Category	Feature	NaiveRAG	LightRAG	GraphRAG	PageIndex	Legal-Onto*	Ours
Retrieval	Semantic Retrieval	✓	✓	–	–	–	✓
	Graph Traversal	–	✓	✓	–	–	✓
	Tree Navigation	–	–	–	✓	–	✓
Knowledge	Knowledge Graph	–	✓	✓	–	✓	✓
	Domain Ontology	–	–	–	–	✓	✓
Reasoning	Multi-hop	–	–	✓	–	✓	✓
	Intent-aware	–	–	–	–	–	✓
Integration	Multi-tier	–	–	–	–	–	✓
	Rank Fusion	–	–	–	–	–	✓
	Multi-domain	–	–	–	–	–	✓

3. Methodology

3.1. Problem Formulation

Given a legal corpus $\mathcal{D} = \{d_1, \dots, d_m\}$ with hierarchically organized articles and a query q , retrieve a ranked list $A^* = [a_1^*, \dots, a_k^*]$ of relevant articles. We define $A^* = \Phi(q, \mathcal{M})$ where $\mathcal{M} = (\mathcal{O}, \mathcal{G}, \mathcal{T}, \Phi)$ integrates ontology $\mathcal{O}$ (terminological), knowledge graph $\mathcal{G}$ (assertional), document forest $\mathcal{T}$ (structural), and retrieval functions Φ .

3.2. System Overview

Figure 1 illustrates the architecture. The **offline phase** constructs $\mathcal{M}$ through parsing, extraction, and ontology generation. The **online phase** processes queries via ontology expansion (Stage 0) followed by three retrieval tiers leveraging $\mathcal{T}$, $\mathcal{G}$, and $\mathcal{O}$.

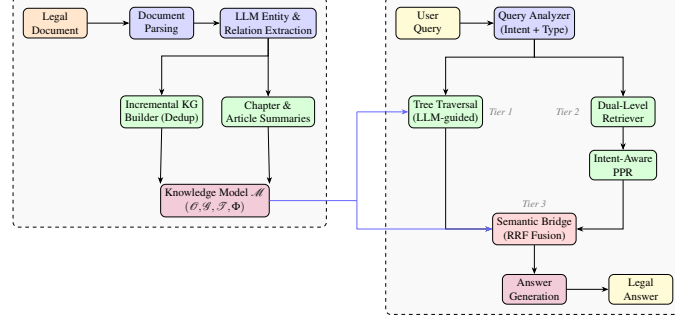


Figure 1. System architecture. Offline: knowledge model construction. Online: three-tier query processing (Tier 1: tree traversal, Tier 2: semantic+PPR, Tier 3: RRF fusion).

3.3. Knowledge Representation Model

Our model extends Legal-Onto [8], which defines $K = (C, R, RULES) + (Key, Rel, weight)$ integrating the Rela-model [21] with conceptual graphs. We extend this by separating terminological from assertional knowledge, adding structural representation, and replacing manual rules with LLM-based retrieval.

Definition 1 (Legal Knowledge Model). *Given a legal corpus $\mathcal{D}$, the knowledge model is:*

$$\mathcal{M} = (\mathcal{O}, \mathcal{G}, \mathcal{T}, \Phi) \quad (1)$$

where $\mathcal{O}$ is a domain ontology, $\mathcal{G}$ is a knowledge graph, $\mathcal{T}$ is a document forest, and Φ is a set of retrieval functions.

Ontology $\mathcal{O} = (C, P, H, L)$ represents terminological knowledge: C = OWL classes for legal entity types, $P = P_{\text{obj}} \cup P_{\text{data}}$ = object and datatype properties, $H \subseteq C \times C$ = class hierarchy via `rdfs:subClassOf` (4-level taxonomy), and $L : C \cup P \rightarrow \text{Vietnamese}$ = label mapping (e.g., `JointStockCompany` $\mapsto$ “cong ty co phan”). This separates the TBox from Legal-Onto’s flat concept set, enabling class hierarchy traversal for query expansion.

Knowledge Graph $\mathcal{G} = (E, R)$ represents assertional knowledge. Each entity $e = (id, name, type, conf, src)$ carries one of 14 types (Table 3), a confidence score, and source provenance. Each relation $r = (e_i, pred, e_j, evid, conf)$ spans 56 predicate types with evidence text; selected predicates are declared symmetric or transitive to support inference during PPR traversal.

Example: $\mathcal{O}$ vs. $\mathcal{G}$. For the query “Does a joint-stock company need a power-of-attorney when establishing an LLC subsidiary?”, $\mathcal{O}$ recognizes `JointStockCompany` and `LLC_SingleMember` as sibling subclasses of `Company` (H -traversal), expanding the query to match articles governing both types. $\mathcal{G}$ then links entity *nguo-i-dai-dien-phan-von-gop* (authorized representative) via *governed_by* to Art. 79 (delegation rules). In short, $\mathcal{O}$ answers “what kind of entity?” (taxonomy) while $\mathcal{G}$ answers “what rules apply?” (legal facts). All four baselines score $\text{Hit@30} = 0$ on this query; our system ranks Art. 79 first ($\text{MRR} = 1.0$).

Document Forest $\mathcal{T} = \{T_1, \dots, T_m\} \cup X$ represents structural knowledge. Each T_i is a rooted tree with six levels (Document $\rightarrow$ Chapter $\rightarrow$ Section $\rightarrow$ Article $\rightarrow$ Clause $\rightarrow$

Point), and X contains cross-reference edges between documents. The forest spans multiple legal domains, with documents organized into domain groups (e.g., enterprise law, traffic safety law) that enable domain-aware routing during retrieval.

Retrieval Functions $\Phi = \{\phi_0, \phi_1, \phi_2, \phi_3\}$ replace Legal-Onto’s manual inference rules:

- $\phi_0: Q \times \mathcal{O} \rightarrow Q'$ (ontology expansion),
- $\phi_1: Q' \times \mathcal{T} \rightarrow 2^A$ (tree traversal),
- $\phi_2: Q' \times \mathcal{G} \times \mathcal{O} \rightarrow 2^A$ (semantic retrieval),
- $\phi_3: 2^A \times 2^A \times \mathcal{G} \rightarrow A^*$ (fusion).

Where Legal-Onto applies hand-crafted deductive rules, our functions leverage LLM reasoning while preserving formal relation properties in the graph structure.

3.4. Knowledge Graph Construction

The offline phase builds $\mathcal{G}$ in four steps: **(1) Document Parsing** into hierarchical trees forming $\mathcal{T}$; **(2) Entity-Relation Extraction** via one LLM call per article, producing evidence-grounded triples; **(3) Incremental KG Building** with Vietnamese slug normalization (NFD $\rightarrow$ diacritics removal $\rightarrow$ lowercasing, e.g., “Cong ty co phan” $\rightarrow$ *cong-ty-co-phan*) and a 4-pass deduplication pipeline across all 13 documents; **(4) Summary Generation** of chapter and article summaries for Tier 1 traversal. Following knowledge engineering practice in ontology-based expert systems [9], construction is **semi-automatic**: after LLM extraction, retrieval evaluation on a development query set revealed that LLM-generated relations missed most cross-document references. A domain expert then added ~ 130 relations (cross-document and intra-document references, concept entities linking common legal terms to their governing articles). This evaluate-and-refine loop is standard in expert systems and contributes to the retrieval accuracy reported in Section 4.

3.4.1. Intent-Aware Edge Weights

Table 2 defines intent-specific weight multipliers for PPR.

Table 2. Intent-relation weight matrix.

Relation	Definition	Procedure	Right	Obligation	Penalty
defines_as	1.0	0.5	0.5	0.5	0.5
includes	0.95	0.5	0.5	0.5	0.5
requires	0.5	1.0	0.5	0.85	0.5
has_authority	0.5	0.5	1.0	0.5	0.5
has_obligation	0.5	0.5	0.5	1.0	0.5
has_penalty	0.5	0.5	0.5	0.5	1.0

This allows PPR to traverse paths most relevant to query intent (e.g., *defines_as* for definitions, *requires* for procedures).

3.5. Three-Tier Retrieval

3.5.1. Stage 0: Query Analysis

The query analyzer performs: (1) **intent detection** across 6 types (Penalty, Definition, Procedure, Requirement, Reference, General) via Vietnamese regex with LLM fallback; (2) **type classification** into 7 categories; (3) **abbreviation expansion** (e.g., *TNHH* $\rightarrow$ *trach nhien huu han*); and (4) **ontology expansion** $\phi_0(q, \mathcal{O})$ via class hierarchy traversal in H with Vietnamese label lookup through L .

3.5.2. Tier 1: LLM-Guided Tree Traversal

Tier 1 navigates $\mathcal{T}$ using LLM reasoning in three loops: **Loop 0** (Forest→Domain→Document) first matches the query to a domain group using pre-computed domain keywords, then selects top 5 documents within the matched domain; **Loop 1** (Document→Chapter) selects top 8 chapters using summaries and keywords; **Loop 2** (Chapter→Article) identifies top 7 articles from article summaries. General Provisions chapters are automatically included. Each loop produces a reasoning path for interpretable retrieval.

3.5.3. Tier 2: Dual-Level Semantic Retrieval

Tier 2 performs hybrid semantic search over $\mathcal{G}$ and $\mathcal{O}$ with six components. *Low-level* (entity-specific): (1) keyphrase matching, (2) query-entity cosine similarity, (3) intent-aware PPR, (4) ontology class matching. *High-level* (theme/concept): (5) legal theme identification, (6) hierarchy traversal through $\mathcal{O}$'s taxonomy. The combined score is:

$$S_{T2} = .05S_{kp} + .20S_{con} + .25S_{PPR} + .20S_{sem} + .15S_{thm} + .15S_{hier} \quad (2)$$

Multi-Intent Personalized PageRank. The PPR component computes query-specific node importance on $\mathcal{G}$:

$$PPR(v) = (1 - \alpha) \sum_{u \in N(v)} \frac{w(u, v) \cdot PPR(u)}{d(u)} + \alpha \cdot p(v) \quad (3)$$

where $\alpha = 0.15$ is the teleport probability, $w(u, v)$ is the edge weight adjusted by query intent (Table 2), $d(u)$ is the weighted out-degree of u , and $p(v)$ is the personalization vector seeded from matched entities.

3.5.4. Tier 3: Semantic Bridge (RRF Fusion)

Tier 3 merges results from Tier 1 and Tier 2 using Reciprocal Rank Fusion [28]:

$$RRF(d) = \sum_{s \in \{T1, T2, KG\}} w_s \cdot \frac{1}{k + \text{rank}_s(d)} \quad (4)$$

where $k = 60$ and source weights are $w_{T1} = 1.4$, $w_{T2} = 0.7$, $w_{KG} = 1.0$. Tree traversal receives the highest weight as it provides the most reliable structural signal; KG expansion supplements with cross-reference articles outside the traversal path.

Knowledge graph expansion. After initial fusion, we expand results through $\mathcal{G}$'s relation edges. Strong relations (*references*, *requires*, *depends_on*) are prioritized for cross-chapter linking (up to 5 articles), while expansion relations (*defined_as*, *related_to*) handle same-chapter augmentation (up to 2 articles). Cross-document references are explicitly resolved via document-prefixed entity IDs.

Adjacent article expansion. After RRF scoring, we expand ± 2 articles from each result with decay factor 0.7, as correct answers often lie in neighboring articles. Top-ranked articles are passed to an LLM for answer generation.

4. Experiments

4.1. Experimental Setup

Dataset. 13 Vietnamese legal documents across two domains: enterprise law (Law 59/2020/QH14 + 9 decrees) and traffic safety law (Law 36/2024/QH15 + 2 penalty decrees), totaling

77 chapters and 840 articles. Evaluation set: 200 expert-annotated questions (150 enterprise, 50 traffic; 1–8 relevant articles each, avg. 2.3) from legal platforms LuatVietnam and ThuVienPhapLuat.

Knowledge graph statistics. Table 3 summarizes the constructed knowledge graph.

Table 3. Knowledge graph statistics.

Entity Type	Count	Relation Type	Count	Structure	Count
Total Entities	5,936	Total Relations	5,633	Document Tree Nodes	9,226
Action	1,903	requires	1,377	Documents	13
Legal Term	1,632	applies_to	810	Chapter Summaries	77
Organization	500	has_authority	506	Article Summaries	840
Legal Reference	451	references	434	Cross-references	168
Person/Role	392	condition_for	361		
Penalty	327	includes	358		
Other (8 types)	731	Other (50 types)	1,787		

Implementation. Claude 3.5 Haiku for tree traversal, query analysis and entity extraction. Embeddings: paraphrase-multilingual-MiniLM-L12-v2 [2]. PPR: iterative computation, max 50 iterations, tolerance 10^{-6} . All LLM responses cached in SQLite (~ 2.1 GB) for reproducibility.

Metrics. Hit@ k , Recall@ k , Precision@ k , NDCG@ k at $k \in \{1, 3, 5, 10, 20, 30\}$, and MRR.

4.2. Baselines

BM25 [18] ($k_1 = 1.2$, $b = 0.75$); **TF-IDF** [19] with Vietnamese segmentation; **Semantic Search** [2] using multilingual sentence-transformers; **LightRAG** [5] with default parameters; **PageIndex** [7] (our Tier 1 in isolation). All baselines retrieve from the full 13-document corpus with max_results=30.

4.3. Main Results

Table 4 presents Hit@ k performance.

Table 4. Hit@ k performance (%). Best **bold**, 2nd underlined.

Method	Hit@5	Hit@10	Hit@20	Hit@30	MRR
Semantic	10.0	15.5	24.0	25.5	0.144
Keyword	22.0	33.0	51.5	64.0	0.161
BM25	21.0	36.0	58.5	67.5	0.144
TF-IDF	<u>26.0</u>	<u>40.5</u>	<u>59.5</u>	<u>71.0</u>	<u>0.191</u>
3-Tier (ours)	65.0	84.5	87.5	90.5	0.461

Our 3-Tier system reaches 84.5% Hit@10 (+44.0pp over TF-IDF, the best baseline), with Recall@10 = 55.2% and NDCG@10 = 41.3%. Semantic search (15.5%) fails on Vietnamese legal terminology; BM25/TF-IDF (~ 36 –41%) lack structural awareness.

4.4. Multi-Domain Performance

Table 5 compares per-domain performance.

Table 5. Per-domain Hit@ k (%) for 3-Tier system.

Domain	n	Hit@1	Hit@5	Hit@10	Hit@20	Hit@30	MRR
Enterprise Law	150	29.3	62.7	83.3	86.7	88.7	0.436
Traffic Safety Law	50	38.0	72.0	88.0	90.0	96.0	0.535
Combined	200	31.5	65.0	84.5	87.5	90.5	0.461

Traffic safety law scores higher (88.0% vs. 83.3% Hit@10) because its penalty articles map directly to specific violations, while enterprise law involves more cross-decree references. No domain-specific tuning was applied, identical pipeline, parameters, and RRF weights serve both domains.

4.5. Analysis

4.5.1. Performance by Query Category

Table 6 breaks down Hit@10 by question category.

Table 6. Hit@10 by question category (%).

Category	Hit@10	Count
<i>Enterprise Law</i>		
Capital contributions	100.0	18
Share/capital transfer	100.0	11
Dissolution & bankruptcy	95.7	47
Legal representatives	83.3	6
Business registration changes	78.6	28
Company formation	69.2	13
Company types	69.2	13
Other enterprise	42.9	14
<i>Traffic Safety Law</i>		
Traffic administrative procedures	92.0	25
Traffic violation penalties	91.7	12
Traffic rules	75.0	8
Other traffic	80.0	5

Capital-related queries (100%) benefit from Tier 1’s direct navigation; traffic penalties (91.7%) from PPR’s *has_penalty* weight. Lower-performing categories (company formation/types, 69.2%) involve cross-decree references where relevant articles reside outside the primary law.

4.5.2. Qualitative and Error Analysis

Success (multi-hop): “If a member borrows money to meet charter capital and later withdraws it, what are the penalties?” (Tier 1 navigates to Article 47 on member obligations; Tier 2’s PPR follows *requires* edges from the “charter capital” entity to Article 16, Clause 5 listing prohibited acts including fraudulent capital declaration; Tier 3 fuses both in top 5, achieving **Hit@5**). **Success (cross-chapter):** “A partnership member contributed a house worth 1 billion VND; it is now worth 1.5 billion, can they withdraw it?” (Tier 1 reaches Articles 178–180 on partnership rules; Tier 2 adds Article 186 on capital transfer via *condition_for* edges, correctly identifying that withdrawal is pro-

hibited but transfer is permitted; **Hit@5**. **Failure:** “Requirements for foreign-invested enterprise capital structure?” (requires synthesizing Investment Law articles not in our corpus; Tier 2 partially reaches related articles but ranks at position 15).

Among the 31 failures (15.5%): **cross-corpus gaps** (12, 38.7%)—answers in laws outside our corpus; **compound term splitting** (9, 29.0%)—Vietnamese segmentation fragments legal terms; **ambiguous intent** (10, 32.3%)—mixed intents bias PPR traversal. All LLM responses are cached in SQLite (~2.1 GB) for deterministic reproducibility.

5. Conclusion

This paper introduced a three-tier knowledge graph RAG system for Vietnamese legal QA, built on a formal knowledge model $\mathcal{M} = (\mathcal{C}, \mathcal{G}, \mathcal{T}, \Phi)$ that extends Legal-Onto by separating terminological, assertional, and structural representations. On 200 real questions spanning two legal domains, the system reaches 84.5% Hit@10 (MRR 0.461), improving over the strongest baseline by 44.0pp.

The central finding is that structural navigation and semantic graph retrieval are complementary: Tier 1’s tree traversal provides the most reliable signal for Vietnamese legal documents (hence $w_{T1} = 1.4$), while Tier 2’s intent-aware PPR catches cross-reference articles that lie outside the traversal path. This complementarity holds across domains: 83.3% Hit@10 on enterprise law and 88.0% on traffic safety law, despite substantial structural differences between the two domains and no domain-specific tuning. Adding a new legal domain requires only parsing documents and running the same KG construction pipeline.

Limitations. Tier 1 requires multiple LLM calls (~7 s latency per query). Entity extraction lacks gold-standard validation. The two domains tested share Vietnamese legal formatting; generalization to other legal traditions remains untested.

Future work. We plan to extend coverage to Investment, Tax, and Labor law, add multi-turn conversational support, and produce explainable retrieval traces citing graph paths used in ranking.

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